

Luke Liang

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EDUCATION

MS in Computer Science, Machine Learning Track Expected Dec 2026
Columbia University New York, NY

BS in Computer Science, BS in Data Science & Statistics, Math Minor (GPA: 3.86) Aug 2021 - May 2024
Miami University Oxford, OH

- Astronaut Scholarship (*national*), Choose Ohio First ML Scholarship, Summa Cum Laude, Honors College
- Trained at University of Tennessee (*National Science Foundation sponsored*) for ML and data science through the HPC at Oak Ridge national laboratory (*only student selected*)
- Coursework: Statistical Learning, Regression Analysis, Bayesian Statistics, Analysis of Forecasting Systems, Real Analysis, Quantum Computing, Algorithms, Systems 2, High Performance Computing, Machine Learning, Network Analysis and Modeling, Predictive Models for Health

PROFESSIONAL EXPERIENCE

Research Assistant Sep 2025 - Present
Columbia University New York, NY

- Working in an epidemiology lab with a focus on data analysis and machine learning

Research Assistant Aug 2021 - Oct 2024
Miami University Oxford, OH

- Published 4 papers focused on ML and network science (*1 first author, 3 co-author*). Two first author manuscripts involving web applications and simulation under preparation
- Created a self-contained web-app project housing a simulation platform for researchers at the Centers for Disease Control and Prevention (CDC) to evaluate policy and intervention effectiveness on youth suicide prevention (**Python, Numba, NumPy, Docker, HPC**)
- Applied various ML models and several methods to augment image data to identify erroneous network simulation models with up to 95% accuracy in select cases (**Python, ML, TensorFlow, NumPy, pandas**)
- Developed a cell-to-cell HIV simulation with mutations and drug resistance for treatments (**Python, NumPy**)
- Drafted a National Health Institute suicide modeling grant proposal

Teaching Assistant (Machine Learning & Algorithms) Aug 2022 - Dec 2023
Miami University Oxford, OH

- Led and published an extension of a previous research project by exploring data compression techniques and several ML models for identifying erroneous network simulations (**Python, ML**)
- Tutored students in Python, data structures, algorithms, and machine learning
- Ran several weekly Python bootcamps for students across all disciplines
- Provided feedback and graded exams, assignments, quizzes, and projects for students

Software Engineering Intern Jun 2022 - Aug 2022
JPMorgan Chase & Co. Chicago, IL

- Migrated internal software from AngularJS to ReactJS and deployed infrastructure (**ReactJS, Jenkins**)
- Won the Code for Good Hackathon 2022 with a web-based project to revamp a non-profit organization's website with a group of other interns (**Javascript, MySQL**)
- Integrated legacy pages with new internally developed APIs and updated technical documents

LEADERSHIP EXPERIENCE

Chapter President May 2023 - May 2024
Tau Beta Pi - The Engineering Honor Society Oxford, OH

- Established chapter through student government to secure funding for social and service events
- Increased membership and community engagement by over 70% after inactivity due to COVID

PUBLICATIONS

Experimental evaluation of a machine learning approach to improve the reproducibility of network simulations

L. Liang, H. Phan, P. J. Giabbanelli

Simulation, 2024, ([link](#))

Simulation models for suicide prevention: a survey of the state-of-the-art

R. Schuerkamp, **L. Liang**, K.L. Rice, P. J. Giabbanelli

Computer 12.7, 2023, ([link](#))

Preserving simulation insight while removing data: verification of compressed simulation traces via machine learning

M. Nguyen, D. Vu, A. Vo, **L. Liang**, P. J. Giabbanelli

Annual Modeling and Simulation Conference (ANNSIM), 2023, ([link](#))

A new application of machine learning: detecting errors in network simulations

M. K. Wozniak, **L. Liang**, H. Phan, P. J. Giabbanelli

Winter Simulation Conference (WSC), 2022, ([link](#))

MANUSCRIPTS

A cellular automata approach to evaluating drug effectiveness for HIV

L. Liang, Z. Sun, P. J. Giabbanelli

In preparation

An artificial intelligence approach to support youth suicide prevention initiatives in the U.S.A

L. Liang, R. Schuerkamp, K. L. Rice, M. M. Brown, N. Nataraj, J. Mendoza-Alonzo, C. R. Harper, C. Florence, B. D. Maldonado, B. Moran, P. J. Giabbanelli

In preparation

Impact of policies, practices, and programs on preventing adolescent suicide: an agent-based modeling approach

K. L. Rice, P. J. Giabbanelli, M. M. Brown, N. Nataraj, C. R. Harper, **L. Liang**, R. Schuerkamp, C. Florence

In preparation

PRESENTATIONS

Presenting author listed first

Oral Presentation. **L. Liang**, M. Nguyen, D. Vu, A. Vo, P. J. Giabbanelli, *Preserving simulation insight while removing data: verification of compressed simulation traces via machine learning*, ANNSIM conference. May 2023

Oral Presentation. **L. Liang**, M. K. Wozniak, H. Phan, P. J. Giabbanelli, *A new application of machine learning: detecting errors in network simulations*, WSC online. Dec 2022.

Poster Presentation. **L. Liang**, M. K. Wozniak, H. Phan, P. J. Giabbanelli, *A new application of machine learning: detecting errors in network simulations*, COF Scholar showcase online. Feb 2022.